

CRM Manual Production Validation Plan

Operational Testing Matrix for Real Development Database & Tenants

Engine: Antigravity v2.5 | Lead Schema: 7 Fields | Passive Side-Effect Invariant Enforced

SECTION 1: How to Prepare for Manual Testing

1. Launch the Chatbot Dev Control Center: <http://localhost:3000/dev>
2. Select the targeted Tenant and Account from the top-bar dropdown selectors.
3. Always use a fresh recommended customer ID (e.g. crm-manual-ecom-01) to ensure clean history.
4. Do NOT click "Reset Session" during flow verification; use "Open Conversation" to inspect history.
5. Switch to the "CRM (Leads)" tab to verify lead generation, status updates, and account scoping.

SECTION 2: Best Tenant / Account for Each Scenario

[] Primary Production-Like Ecommerce Tenant: AnimeVerse

Tenant ID:	animeverse
Account ID:	animeverse-store (AnimeVerse Store)
Catalog:	Moon Ninja Hoodie (399 MAD), Neon Ronin T-Shirt (249 MAD), Cyber Spirit Jacket (599 MAD)
Capabilities:	Ecommerce enabled, Multilingual Purchase Intent (EN, FR, AR, Darija, Arabizi)
Best Use:	Ecommerce browsing, price/stock queries, BUY_INTENT, media requests, idempotency

[] Primary Consultation / Booking Tenant: Legal Consultation Demo

Tenant ID:	consultation-demo (or test-prec-tenant)
Account ID:	law-firm-main (or test-prec-account)
Workflows:	consultation_booking (Collects name, phone, email, topic, preferred date & time)
Best Use:	Testing sales workflow completion triggering Lead creation (Status: NEW)

[] Primary Operational / Triage Tenant: Dev Control Center

Tenant ID:	dev-tenant
Account ID:	dev-account
Workflows:	feedback_intake_workflow (Operational), triage_workflow (3-branch selector)
Best Use:	Testing feedback zero-lead creation and triage branch discrimination

SECTION 3: Ecommerce & Catalog Intent Tests

[] TC-ECOM-01: Catalog Search & Browsing

Tenant / Account:	animeverse / animeverse-store
Customer ID:	crm-manual-ecom-01
Input:	Show me the hoodie
Expected Intent:	PRODUCT_SEARCH (Resolves ANV-H001 Moon Ninja Hoodie)
Expected CRM:	NO LEAD (Lead Count = 0)
Verify:	Product card rendered; conversation status remains ACTIVE

[] TC-ECOM-02: Price & Availability Inquiries

Tenant / Account:	animeverse / animeverse-store
Customer ID:	crm-manual-ecom-01
Input:	How much is it and is it in stock?
Expected Intent:	PRICE / AVAILABILITY (399 MAD, 25 in stock)
Expected CRM:	NO LEAD (Lead Count = 0)
Verify:	Price answered without creating premature sales lead

[] TC-ECOM-03: Media Request (Image / Video)

Tenant / Account:	animeverse / animeverse-store
Customer ID:	crm-manual-ecom-01
Input:	Show me a picture of it
Expected Intent:	PRODUCT_DETAIL (Media: image)
Expected CRM:	NO LEAD (Lead Count = 0)
Verify:	Media attached to context without triggering sales pipeline

[] TC-ECOM-04: Strong Purchase Intent (English BUY_INTENT)

Tenant / Account: animeverse / animeverse-store
Customer ID: crm-manual-ecom-01
Input: I want to buy this
Expected Intent: BUY_INTENT (Associated with ANV-H001)
Expected CRM: LEAD CREATED (Lead Count = 1, Status = NEW)
Verify: Appears immediately in CRM tab under animeverse-store

SECTION 4: Multilingual Purchase Intent Matrix**[] TC-LANG-01: French Purchase Intent**

Tenant / Account: animeverse / animeverse-store
Customer ID: crm-manual-lang-fr
Input: Je veux acheter le Moon Ninja Hoodie
Expected Intent: BUY_INTENT
Expected CRM: LEAD CREATED (Lead Count = 1, Status = NEW)
Verify: 0 LLM calls for deterministic intent gating

[] TC-LANG-02: Moroccan Darija (Arabic Script Phrase)

Tenant / Account: animeverse / animeverse-store
Customer ID: crm-manual-lang-dar-ar
Input: bghit nchri hadchi (Darija: I want to buy this)
Expected Intent: BUY_INTENT
Expected CRM: LEAD CREATED (Lead Count = 1, Status = NEW)
Verify: Deterministic regex purchase phrase resolution

[] TC-LANG-03: Moroccan Darija (Arabizi Script)

Tenant / Account: animeverse / animeverse-store
Customer ID: crm-manual-lang-dar-lat
Input: bghit nchri had lhoodie
Expected Intent: BUY_INTENT
Expected CRM: LEAD CREATED (Lead Count = 1, Status = NEW)
Verify: Lead created with externalId crm-manual-lang-dar-lat

[] TC-LANG-04: Standard Arabic (MSA Phrase)

Tenant / Account: animeverse / animeverse-store
Customer ID: crm-manual-lang-msa
Input: Uridu shiraa hadha al-muntaj (I want to buy this product)
Expected Intent: BUY_INTENT
Expected CRM: LEAD CREATED (Lead Count = 1, Status = NEW)
Verify: Clean Arabic normalization and lead creation

SECTION 5: Negative Purchase Collision Prevention**[] TC-NEG-01: Recommendation Collision Guard**

Tenant / Account: animeverse / animeverse-store
Customer ID: crm-manual-neg-01
Input: Which jacket should I buy? / achno nchri?
Expected Intent: RECOMMENDATION
Expected CRM: NO LEAD (Lead Count = 0)
Verify: Recommends products without premature lead qualification

[] TC-NEG-02: Order Tracking Query

Tenant / Account: animeverse / animeverse-store
Customer ID: crm-manual-neg-02
Input: Where is my order? / fin talab dyali?
Expected Intent: TRACKING / GENERAL_CONVERSATION
Expected CRM: NO LEAD (Lead Count = 0)
Verify: Tracking queries do not pollute sales CRM

[] TC-NEG-03: Return / Refund Submission

Tenant / Account: animeverse / animeverse-store
 Customer ID: crm-manual-neg-03
 Input: Can I return this item? / bghit nrje3 hadchi
 Expected Intent: RETURNS
 Expected CRM: NO LEAD (Lead Count = 0)
 Verify: Return policy answers delivered without sales signal

SECTION 6: Sales / Booking Workflow Tests**[] TC-WF-01: Consultation Booking Completion**

Tenant / Account: consultation-demo / law-firm-main
 Customer ID: crm-manual-wf-sales
 Turn 1 (Trigger): I want to book a consultation
 Turn 2 (Name): Jane Doe
 Turn 3 (Phone): +1-555-0199
 Turn 4 (Topic): Corporate Tax Law
 Turn 5 (Date): Tomorrow at 10 AM
 Expected CRM: LEAD CREATED (Lead Count = 1, Status = NEW)
 Verify: Session completes with confirmation prompt

SECTION 7: Operational Workflows (Zero Leads)**[] TC-OP-01: User Feedback Intake Survey**

Tenant / Account: dev-tenant / dev-account
 Customer ID: crm-manual-wf-op-01
 Turn 1 (Trigger): feedback_intake_workflow
 Turn 2 (Rating): 5 - Excellent experience
 Expected Outcome: Workflow COMPLETED successfully
 Expected CRM: NO LEAD (Lead Count = 0)
 Verify: Answers stored in WorkflowSession; CRM tab remains empty

[] TC-OP-02: Support Ticket Request

Tenant / Account: dev-tenant / dev-account
 Customer ID: crm-manual-wf-op-02
 Turn 1 (Trigger): support_request
 Turn 2 (Details): Cannot login to customer dashboard
 Expected Outcome: Workflow COMPLETED successfully
 Expected CRM: NO LEAD (Lead Count = 0)
 Verify: Support ticket recorded; CRM tab remains empty

SECTION 8: Combined Ecommerce + Workflow Idempotency**[] TC-COMB-01: BUY_INTENT Followed by Booking Workflow**

Tenant / Account: animeverse / animeverse-store
 Customer ID: crm-manual-combined-01
 Step 1: Send: "I want to buy the Cyber Spirit Jacket" -> Lead 1 Created (NEW)
 Step 2: In CRM UI: Advance Lead Status to QUALIFIED
 Step 3: Send: "consultation_booking" -> Complete booking flow
 Step 4: Send: "feedback_intake_workflow" -> Complete feedback flow
 Expected CRM: EXACTLY 1 LEAD (Status remains QUALIFIED)
 Verify: No duplicate lead created; merchant pipeline status preserved

SECTION 9: Multi-Branch Triage Workflow

[] TC-TRI-01: Triage Sales Path (Pricing Plans)

Tenant / Account: dev-tenant / dev-account
 Customer ID: crm-manual-tri-sales
 Turn 1: Trigger triage_workflow
 Turn 2: Select Option 1: "Pricing Plans"
 Terminal Node: states.plans (Sales outcome)
 Expected CRM: LEAD CREATED (Lead Count = 1, Status = NEW)

[] TC-TRI-02: Triage Support Path (Technical Support)

Tenant / Account: dev-tenant / dev-account
 Customer ID: crm-manual-tri-supp
 Turn 1: Trigger triage_workflow
 Turn 2: Select Option 3: "Technical Support"
 Terminal Node: states.support (Operational outcome)
 Expected CRM: NO LEAD (Lead Count = 0)

[] TC-TRI-03: Triage Refund Path (Refund Request)

Tenant / Account: dev-tenant / dev-account
 Customer ID: crm-manual-tri-ref
 Turn 1: Trigger triage_workflow
 Turn 2: Select Option 2: "Refund Request"
 Terminal Node: states.refunds (Operational outcome)
 Expected CRM: NO LEAD (Lead Count = 0)

SECTION 10 & 11: CRM UI & Status Transition Tests**[] TC-UI-01: Status Transition Lifecycle**

Tenant / Account: animeverse / animeverse-store
 Customer ID: crm-manual-status-01
 Step 1: Create lead via: "I want to buy the Moon Ninja Hoodie"
 Step 2: In CRM UI, click "View / Manage" on the lead
 Step 3: Change Status: NEW -> CONTACTED -> QUALIFIED -> WON
 Step 4: Send another message: "I want to order more"
 Expected CRM: Lead Status remains WON (No regression to NEW)
 Verify: Status dropdown updates reactively via PATCH /api/dev/crm/leads/:id

SECTION 12: Open Conversation Forensic Verification**[] TC-OPEN-01: Read-Only History vs Reset Session**

Tenant / Account: animeverse / animeverse-store
 Customer ID: crm-manual-open-conv
 Step 1: Generate 4 customer turns (browsing, inquiry, buy intent)
 Step 2: Go to CRM tab -> Click "View / Manage" -> Click "Open Conversation"
 Expected Result 1: Chat UI loads exact 4-message history in chronological order
 Expected Result 2: Conversation status remains ACTIVE (No "Conversation archived" alert)
 Step 3 (Reset): Click "Reset Session" button separately in Chat tab
 Expected Result 3: Only THEN does conversation archive and clear UI

SECTION 13: Tenant & Account Isolation Matrix**[] TC-ISO-01: Cross-Tenant Isolation**

Tenant A: animeverse / animeverse-store (Lead created)
 Tenant B: consultation-demo / law-firm-main
 Action: Switch tenant dropdown from animeverse to consultation-demo
 Expected CRM: AnimeVerse lead is completely invisible in consultation-demo

[] TC-ISO-02: Cross-Account Isolation

Tenant:	Tenant-Acceptance-1787911671893 (or dual account tenant)
Account A:	store-alpha (Lead created)
Account B:	store-beta
Action:	Switch account dropdown from store-alpha to store-beta
Expected CRM:	store-alpha lead is completely invisible in store-beta

SECTION 14 & 15: Production Smoke Test Checklist

- [] 1. Ecommerce BUY_INTENT generates Lead (Status: NEW)
- [] 2. Multilingual buy phrases (FR, AR, Darija, Arabizi) generate Lead
- [] 3. Recommendation & policy questions generate 0 Leads
- [] 4. Booking/consultation workflow completion generates Lead
- [] 5. Feedback, support, tracking, returns generate 0 Leads
- [] 6. Triage sales path generates Lead; support/refund paths generate 0 Leads
- [] 7. Arbitrary unknown workflows generate 0 Leads
- [] 8. Combined flows maintain exactly 1 Lead per customer/account
- [] 9. Advanced Lead status (QUALIFIED, WON) is preserved
- [] 10. Open Conversation loads history in read-only mode without resetting
- [] 11. Tenant and account scoping strictly isolates data
- [] 12. CRM errors or timeouts cannot block customer turns